Experiment 3: What the Pose Error Rises With

Insight analysis for Chapter 4, Section 4.3

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Purpose of this document. The raw write-up of Experiment 3 (`insights/experiment_3/`) reports what each run scored. This document asks *why*: whether the pose error rises when the robot turns, when a movable object is in the camera's view, or after the tracker loses features, and what the robot is doing at the moment the error starts to grow. Every question is tested within each run, so that the run is its own control, and the answer is stated with the evidence that supports it and the evidence that is missing. Every number is written by `script/exp3_insights.py` into `output/compare/simulation/exp3-insights/`; the tables below are input from there unchanged.

1 Questions and method

Five questions were put to the 114 Experiment 3 runs of both testbeds (the registry of `script/analyse_exp3_timelines.py`: 64 dynamic and 18 static runs in the extreme world, 26 dynamic and 6 static in the realistic world).

- Q1. Does the error grow faster while the robot turns than on straights?
- Q2. Does it grow faster while a movable object is in the camera's view?
- Q3. Is the onset preceded by a loss of tracked features?
- Q4. What is the robot doing at the onset: turning or straight, object in view or not?
- Q5. Is the error that these questions concern the estimator's error at all, or does part of it come from how the estimate is scored?

Within-run contrasts. The error series is the start-anchored planar error of the timeline analysis: the estimate is aligned to the ground truth by a rigid fit over the first 10–40 s, after the initialisation transient and before the ground truth has turned 60°, and the planar distance to the ground truth is followed from there. For Q1 and Q2 the samples of a run are split into two states and the growth rate of the error (metres per second of positive and negative change) is computed in each. States are: *turning*, ground-truth yaw rate above $10\,^\circ\,\mathrm{s}^{-1}$; *object in view*, for the extreme bags the projected simulator truth of Experiment 2 (image area of moving objects in the left camera), for the realistic bags the carriers and carried loads whose poses the bag records on `/world/default/pose/info`, placed within 20 m and $\pm 45^\circ$ of the ground-truth heading with no shelf footprint on the line of sight. Rates are taken only while the estimate is still tracking: after the anchor window, before its first impossible-speed jump and while the error is below 3 m. A contrast needs at least 3 s in each state; across runs the two rates are compared with the sign count and the Wilcoxon signed-rank test. For Q3 the matched-feature count (ORB-SLAM3 map inliers, VINS-Fusion tracked features) is followed around the onset as a percentage of the run's own level before it. For Q4 the state at the onset time is read off. The onset is the one the raw write-up and the timeline tables use: 1 s of sustained excess over three times the static median error, with the ground truth scored as the chapter scores it. The drift contrasts of Q1 and Q2 are instead computed with the ground truth at the IMU point (Insight 1), because at the model origin they measure the lever arm, not the estimator.

Lever arm (Q5). Both estimators report the pose of the IMU: the ORB-SLAM3 wrapper converts the camera pose with the camera-to-body extrinsic before writing it, and VINS-Fusion writes its body state. The ground truth is the model's odometry, which reports the model origin. In the simulated platform the IMU sits 0.338 m ahead of that origin (`models/ackermann_robot/model.sdf`). No scoring step in the project compensates this; Chapter 3 states the requirement only for the real robot's tracker marker. The effect is tested by moving the ground truth forward along its own heading by a lever arm L and re-scoring every tracked unfiltered run for L from 0–0.60 m.

2 Insight 1: the error that appears at turns is the scoring lever arm

Scored as the chapter scores them, the error of the tracked runs grows while the robot turns and holds on straights; scored at the IMU point, that pattern disappears and the whole-run ATE falls by half or more. Tables I.1 and I.2 and figure I.1 carry the sweep.

Table I.1: Lever-arm sweep. Both estimators report the IMU frame, which sits 0.338 m ahead of the model origin that the ground truth reports (`models/ackermann_robot/model.sdf`); no scoring step compensates this. The table re-scores the tracked unfiltered runs with the ground truth moved forward by L along its heading: whole-run aligned planar ATE (median over runs) and the drift rate while turning and on straights (median). If the error that appears at turns were the lever arm, ATE and the turning rate would reach their minimum near $L = 0.34$ m.

Testbed	Estimator	n	$L = 0.00$ m			$L = 0.17$ m			$L = 0.34$ m			$L = 0.50$ m		
			ATE [m]	turn [mm/s]	str.	ATE [m]	turn [mm/s]	str.	ATE [m]	turn [mm/s]	str.	ATE [m]	turn [mm/s]	str.
extreme	VINS-Fusion	9	0.32	2.2	-2.1	0.19	7.6	-0.7	0.10	5.2	0.6	0.14	-0.6	1.4
extreme	ORB-SLAM3	9	0.37	-8.2	2.1	0.28	-1.0	2.2	0.19	-0.9	4.7	0.22	-8.1	3.6
realistic	VINS-Fusion	4	0.37	-0.0	2.8	0.29	3.6	5.5	0.25	1.9	5.7	0.28	7.0	3.4
realistic	ORB-SLAM3	8	0.32	8.7	-1.2	0.19	5.4	-0.1	0.15	0.0	2.0	0.24	-2.8	2.0

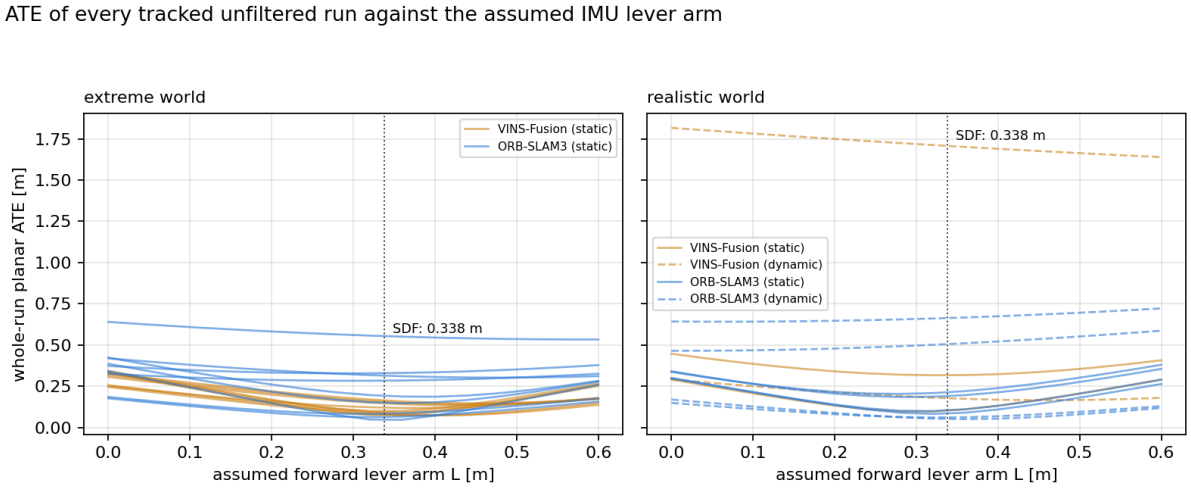


Figure I.1: Whole-run planar ATE of every tracked unfiltered run against the lever arm assumed between the model origin and the IMU. Solid lines are static recordings, dashed the realistic dynamic recordings. The dotted line is the value in the robot model.

The minimum of the ATE lies at the lever arm the robot model specifies. Over the 30 tracked unfiltered runs the ATE-minimising L^* has a median of 0.34 m and an interquartile range of 0.32–0.40 m (table I.2), for both estimators and both testbeds. The three runs that do not follow the pattern are the ones whose error is not a rigid offset: the two ORB-SLAM3 runs on `realistic_002`, whose error is a 0.5–1.0 m step taken in the first 15 s (figure I.4), and the VINS-Fusion run on the same bag, which diverges at its first turn. With $L = 0.34$ m the median static ATE of the extreme world falls from 0.32 m to 0.10 m for

Table I.2: The forward lever arm that minimises each tracked unfiltered run's whole-run planar ATE, from a sweep of L in 0–0.60 m in 0.04 m steps, with the ATE at $L = 0$ (as reported in the chapter) and at the SDF value $L = 0.338$ m. The IMU sits 0.338 m ahead of the model origin in `models/ackermann_robot/model.sdf`.

Testbed	Recording	Estimator	L^* [m]	ATE(0) [m]	ATE(L^*) [m]	ATE(0.338) [m]
extreme	`static_000`	ORB-SLAM3	0.28	0.37	0.33	0.33
extreme	`static_000`	ORB-SLAM3	0.48	0.42	0.30	0.31
extreme	`static_000`	ORB-SLAM3	0.60	0.64	0.53	0.55
extreme	`static_000`	VINS-Fusion	0.44	0.31	0.13	0.15
extreme	`static_000`	VINS-Fusion	0.44	0.32	0.14	0.16
extreme	`static_000`	VINS-Fusion	0.44	0.33	0.15	0.16
extreme	`static_001`	ORB-SLAM3	0.32	0.18	0.06	0.06
extreme	`static_001`	ORB-SLAM3	0.32	0.33	0.28	0.28
extreme	`static_001`	ORB-SLAM3	0.32	0.19	0.08	0.08
extreme	`static_001`	VINS-Fusion	0.40	0.26	0.07	0.08
extreme	`static_001`	VINS-Fusion	0.40	0.25	0.07	0.08
extreme	`static_001`	VINS-Fusion	0.36	0.25	0.12	0.12
extreme	`static_002`	ORB-SLAM3	0.36	0.39	0.15	0.15
extreme	`static_002`	ORB-SLAM3	0.40	0.42	0.19	0.19
extreme	`static_002`	ORB-SLAM3	0.36	0.34	0.05	0.05
extreme	`static_002`	VINS-Fusion	0.32	0.33	0.10	0.10
extreme	`static_002`	VINS-Fusion	0.36	0.34	0.09	0.09
extreme	`static_002`	VINS-Fusion	0.32	0.33	0.07	0.08
realistic	`realistic_000`	ORB-SLAM3	0.28	0.34	0.20	0.21
realistic	`realistic_000`	ORB-SLAM3	0.28	0.34	0.18	0.19
realistic	`realistic_000`	VINS-Fusion	0.32	0.45	0.32	0.32
realistic	`realistic_001`	ORB-SLAM3	0.32	0.30	0.10	0.10
realistic	`realistic_001`	ORB-SLAM3	0.32	0.30	0.08	0.09
realistic	`realistic_001`	VINS-Fusion	0.32	0.29	0.10	0.10
realistic	`realistic_002`	ORB-SLAM3	0.08	0.64	0.64	0.66
realistic	`realistic_002`	ORB-SLAM3	0.00	0.47	0.47	0.51
realistic	`realistic_002`	VINS-Fusion	0.60	1.82	1.64	1.71
realistic	`realistic_003`	ORB-SLAM3	0.32	0.15	0.06	0.06
realistic	`realistic_003`	ORB-SLAM3	0.36	0.17	0.05	0.05
realistic	`realistic_003`	VINS-Fusion	0.48	0.29	0.17	0.18
median L^* = 0.34 m, IQR 0.32–0.40 m over 30 runs						

VINS-Fusion and from 0.37 m to 0.19 m for ORB-SLAM3; in the realistic world from 0.37 m to 0.25 m and from 0.32 m to 0.15 m (table I.1).

The turning effect goes with it. At $L = 0$ ORB-SLAM3's error in the realistic world grows at a median 8.7 mm s^{-1} while turning and shrinks on straights, in eight of eight unfiltered runs; at $L = 0.34$ m the turning rate is 0.0 mm s^{-1}. This is the signature of a rigid offset between the scored point and the reference point: the offset vector rotates with the robot, so the apparent error changes only while the heading changes.

The sweep fits one parameter and could, on its own, absorb some estimator error. What makes it a finding rather than a fit is that the minimum coincides with an independently specified geometric constant, in 30 runs, across two estimators that share no code. The residual ATE at L^* is the estimator's.

[Consequence beyond this experiment] Every simulation ATE in Chapter 4 that aligns an estimate to the model-origin ground truth carries this offset: the static floor of Experiment 3 and the visual estimators of Experiment 1. The wheel and IMU baselines of Experiment 1 are referenced to the same origin and are not affected in the same way, so the estimator ranking of that experiment has to be re-scored before it is quoted. The correction is one line: move the ground truth to the IMU point before alignment. It has since been applied in the scoring scripts (`vio_metrics.load_gt`, 22 October 2026) and every simulation table and figure of Experiments 1 and 3 has been regenerated; the chapter prose that quotes the old values is pending.

3 Insight 2: after the correction, turning does not raise the error

With the ground truth at the IMU point, neither estimator's error grows faster while turning than on straights, in either testbed, and the heading error does not either. Tables I.3 and I.4 give the contrasts; figure I.2 plots each run.

Table I.3: Within-run drift contrasts, extreme world. Each run is its own control: the growth rate of the start-anchored error is computed separately in the two states and the two rates are compared across runs (median of each, number of runs where the first exceeds the second, Wilcoxon signed-rank p). "Turning": ground-truth yaw rate above 10 ° s^{-1}. "Object in view": detector boxes present on the frame (borrowed from the class-filtered run on the same bag for unfiltered runs). Rates are taken only while the estimate is still tracking (before the first jump and below 3 m error). Source: `output/compare/simulation/exp3-insights/insights.csv`.

Condition	Estimator	n	turning	straight	$n_>$	p
			[mm s^{-1}]			
static	VINS-Fusion	9	5.2	0.6	7	0.10
static	ORB-SLAM3	9	-0.8	4.7	1	0.10
dynamic	VINS-Fusion	2	46.4	169.9	0	—
dynamic	VINS-Fusion+YOLO	2	94.0	180.6	0	—
Heading error, same contrast [m° s^{-1}]						
static	VINS-Fusion	9	9.0	9.1	4	0.73
static	ORB-SLAM3	9	-5.5	4.6	3	0.65
dynamic	VINS-Fusion	2	162.5	169.7	1	—
dynamic	VINS-Fusion+YOLO	2	94.6	103.7	0	—
Object in view vs none, dynamic recordings [mm s^{-1}]						
			in view	none		

The static recordings of the extreme world are the only group with enough runs for a test, and the two estimators point opposite ways there: VINS-Fusion is faster while turning in seven of nine runs and ORB-SLAM3 in one of nine, neither at $p < 0.05$. In the realistic world the sign is mixed in every group. The heading-error contrast is flat throughout. On these data the answer to Q1 is no.

Table I.4: Within-run drift contrasts, realistic world. Each run is its own control: the growth rate of the start-anchored error is computed separately in the two states and the two rates are compared across runs (median of each, number of runs where the first exceeds the second, Wilcoxon signed-rank p). "Turning": ground-truth yaw rate above $10\,^\circ\,s^{-1}$. "Object in view": detector boxes present on the frame (borrowed from the class-filtered run on the same bag for unfiltered runs). Rates are taken only while the estimate is still tracking (before the first jump and below 3 m error). Source: `output/compare/simulation/exp3-insights/insights.csv`.

Condition	Estimator	n	turning	straight	$n_>$	p
			[mm s^{-1}]			
static	VINS-Fusion	2	3.9	7.5	1	—
static	ORB-SLAM3	4	0.1	1.6	1	—
dynamic	VINS-Fusion	1	1.8	4.8	0	—
dynamic	ORB-SLAM3	4	-1.3	2.1	2	—
dynamic	VINS-Fusion+YOLO	3	0.1	5.5	1	—
dynamic	ORB-SLAM3+YOLO	9	9.1	4.3	2	0.20
Heading error, same contrast [m° s^{-1}]						
static	VINS-Fusion	2	4.1	3.2	1	—
static	ORB-SLAM3	4	5.6	-6.1	3	—
dynamic	VINS-Fusion	1	7.0	9.4	0	—
dynamic	ORB-SLAM3	4	-15.0	5.8	0	—
dynamic	VINS-Fusion+YOLO	3	0.6	0.1	1	—
dynamic	ORB-SLAM3+YOLO	9	-5.4	3.7	3	0.30
Object in view vs none, dynamic recordings [mm s^{-1}]						
			in view	none		
dynamic	VINS-Fusion+YOLO	1	1.0	4.2	0	—
dynamic	ORB-SLAM3+YOLO	3	14.4	7.0	1	—

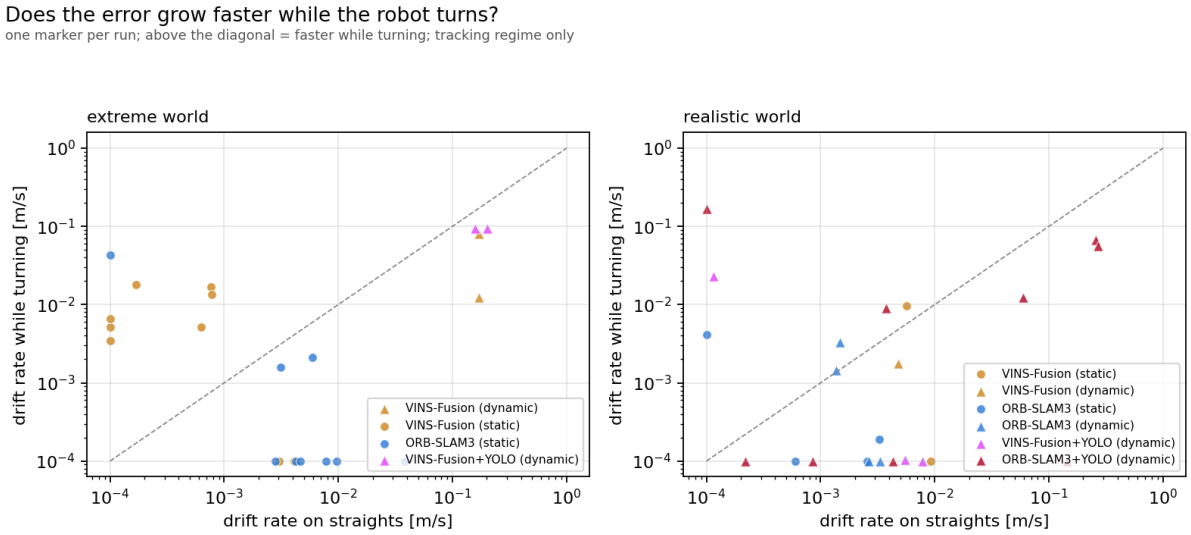


Figure I.2: Drift rate while turning against drift rate on straights, one marker per run, ground truth at the IMU point. Negative rates are drawn at the axis floor. A run above the diagonal drifts faster while turning.

4 Insight 3: in the extreme world the failure is immediate and is not modulated by how much moving object is in view

Every dynamic run of the extreme world passes its onset threshold on the first aisle, almost never while turning, and the moment-to-moment amount of moving object in the image does not predict when ORB-SLAM3 loses tracking or when either estimate jumps. Tables I.5 and I.6 and figure I.3 carry the evidence.

Table I.5: What the robot was doing at the error onset, extreme world, per estimator and condition. "Turning" and "object in view" as in the contrast table; "features" is the median tracked-feature level in the 5 s before onset as a percentage of the run's own level before that. Runs without an onset are counted separately. Source: `output/compare/simulation/exp3-insights/insights.csv`.

Condition	Estimator	runs	no onset	turning [of onsets]	object in view [of onsets]	features [% of own]
static	VINS-Fusion	9	9	0/0	—	—
static	ORB-SLAM3	9	6	1/3	—	100
dynamic	VINS-Fusion	12	0	4/12	9/9	97
dynamic	ORB-SLAM3	12	0	0/12	6/6	—
dynamic	VINS-Fusion+YOLO	20	0	3/20	17/17	87
dynamic	ORB-SLAM3+YOLO	20	0	8/20	16/16	89

Table I.6: Extreme world, dynamic recordings: tracking events against the *true* moving-object area in view, from the projected simulator truth of Experiment 2 (`output/yolo_eval/dyn_*/yolo_eval.csv`). Within each run the frames are split into terciles of moving-object area; the table gives the median over runs of the share of frames in which ORB-SLAM3 was not in the OK state, the rate of impossible-speed jumps of the estimate, and the Spearman correlation between matched features (ORB-SLAM3 map inliers, VINS-Fusion tracked features) and moving-object area. $n_>$ counts runs where the top tercile exceeds the bottom; p is the Wilcoxon signed-rank test over runs.

Estimator	n	area med. [%]	top tercile	bottom tercile	$n_>$	p
Share of frames not tracking OK						
ORB-SLAM3	12	5.7	0.195	0.147	5	0.79
Share of frames not tracking OK						
ORB-SLAM3+YOLO	20	5.5	0.132	0.084	10	0.39
Jumps of the estimate per second						
VINS-Fusion	12	5.5	21.65	20.94	7	0.03
ORB-SLAM3	12	5.7	7.04	5.34	6	0.97
VINS-Fusion+YOLO	19	5.4	20.62	18.18	12	0.10
ORB-SLAM3+YOLO	20	5.5	0.70	1.22	10	0.52
Spearman ρ, matched features against moving-object area (median over runs; runs with $\rho < 0$)						
VINS-Fusion	12		ρ = 0.19		3	
ORB-SLAM3	12		ρ = -0.11		11	
VINS-Fusion+YOLO	19		ρ = 0.18		7	
ORB-SLAM3+YOLO	20		ρ = -0.11		14	

Moving objects are in view for 74–91 % of the frames of these four bags (projected truth of Experiment 2), so a within-bag "no object" baseline does not exist; the comparison has to be between more and less moving area. Split into terciles of moving area within each run, the share of frames in which ORB-SLAM3 is not tracking OK is not higher in the top tercile than in the bottom one (table I.6), nor is the jump rate of either estimate. The matched-feature count falls weakly with moving area (median Spearman ρ about -0.1 for ORB-SLAM3, in eleven of twelve runs).

The onset is on a straight. Of the 64 dynamic runs, the robot was turning at onset in fifteen (table I.5); the first heading change of every route comes after the first aisle, and the error has passed threshold before

What precedes the error onset

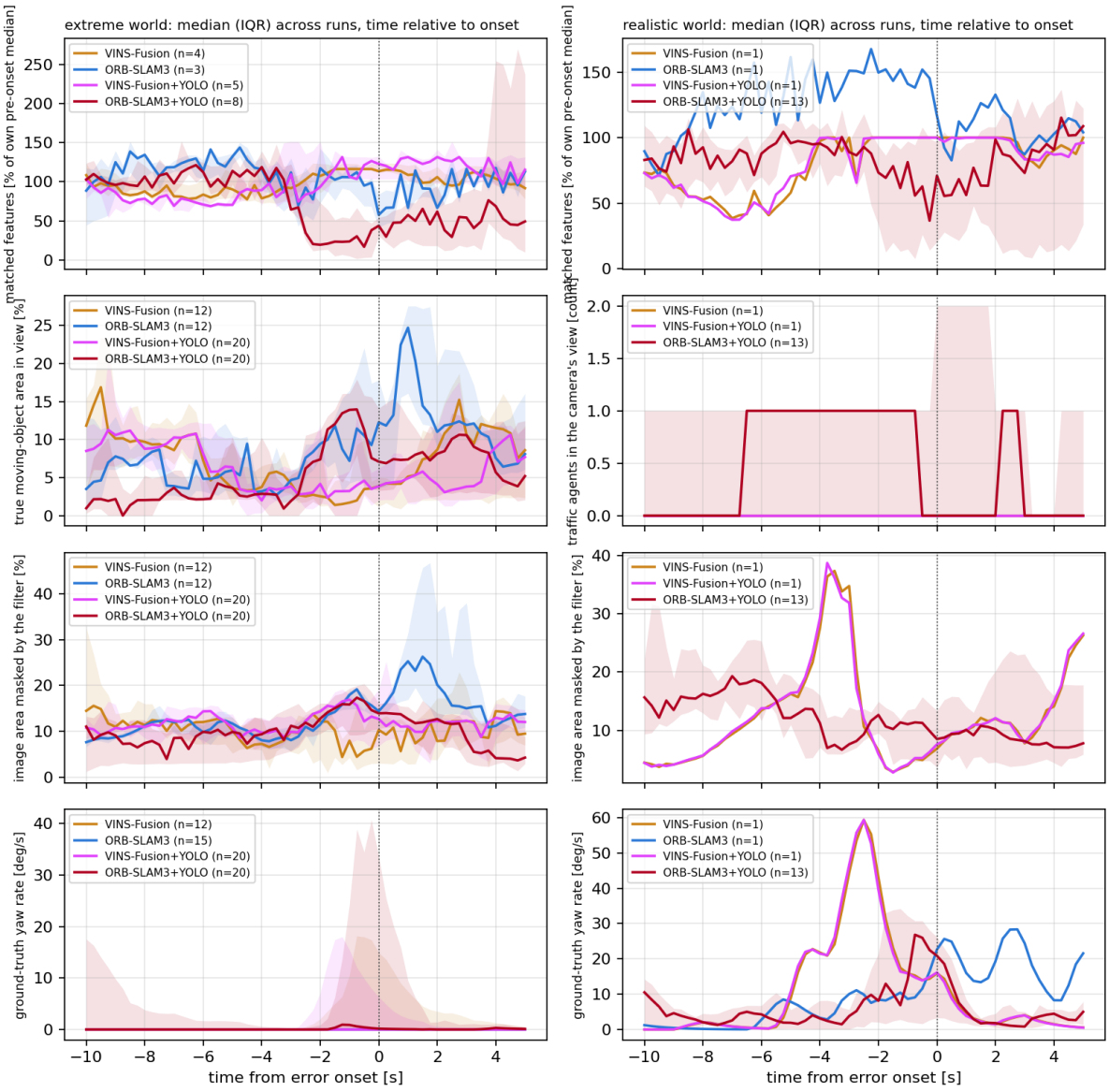


Figure I.3: What precedes the onset, at the chapter's onset definition: median over runs (shaded, interquartile range) of the matched-feature count relative to the run's own pre-onset level, the true moving-object area in view (extreme world) or the number of traffic agents in view (realistic world), the image area the filter masked, and the ground-truth yaw rate, against time from onset.

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The two filtered estimators reach their onset differently. For ORB-SLAM3 with the class mask the matched features collapse to about one fifth of the run's own level 2.5 s before the onset (figure I.3, top left): the mask removes the features the tracker was using, and the error follows. VINS-Fusion shows no feature loss before its onset, filtered or not; its estimate does not drift, it breaks, with a jump rate of about 20 per second once it has failed. What breaks it is not identified by these data; the hypothesis that wrongly tracked moving points corrupt the visual-inertial initialisation remains untested.

5 Insight 4: in the realistic world the traffic in view does not measurably change the drift, and the class mask costs features when traffic is in view

With the carriers' true positions from the bag, the drift of the tracked realistic runs is not consistently faster while a carrier is in the camera's view, and ORB-SLAM3 with the class mask has fewer matched features during those moments without a drift penalty. Table I.7 and figure I.4 carry the evidence; the per-run values are in `output/compare/simulation/exp3-insights/realistic_runs_lever0p338.tex`.

Table I.7: Realistic world, dynamic recordings: within-run contrasts between the moments a traffic agent (carrier or person) is in the camera's view and the moments none is. Visibility is computed from the bag's own model poses (`/world/default/pose/info`): within 20 m, within ±45° of the ground-truth heading, and with no shelf footprint on the line of sight. Drift rate while tracking (before the first jump, below 3 m); share of frames ORB-SLAM3 was not tracking OK; median matched features (ORB-SLAM3 map inliers, VINS-Fusion tracked features); jumps per second. $n_>$ counts runs where the in-view value exceeds the other; p is the Wilcoxon signed-rank test.

Quantity	Estimator	n	in view	none	$n_>$	p
drift rate [mm s^{-1}]	VINS-Fusion	1	2.0	5.7	0	—
	ORB-SLAM3	4	-0.2	1.5	1	—
	VINS-Fusion+YOLO	3	2.4	9.1	0	—
	ORB-SLAM3+YOLO	10	42.0	8.2	7	0.08
not tracking OK [share]	ORB-SLAM3	4	0.000	0.000	0	—
	ORB-SLAM3+YOLO	16	0.003	0.029	5	0.53
matched features	VINS-Fusion	2	150	150	0	—
	ORB-SLAM3	4	431	453	2	—
	VINS-Fusion+YOLO	4	150	150	0	—
	ORB-SLAM3+YOLO	16	350	370	5	0.05
jumps [s^{-1}]	VINS-Fusion	2	0.00	0.00	0	—
	ORB-SLAM3	4	0.00	0.01	0	—
	VINS-Fusion+YOLO	4	0.00	0.00	0	—
	ORB-SLAM3+YOLO	16	0.05	0.07	7	0.82

Traffic was in view for a median 37 % of the tracked time.

Traffic is in view for a median 37 % of the tracked time, so both states are well populated; the groups of runs are not. Two unfiltered VINS-Fusion runs and four ORB-SLAM3 runs carry the unfiltered answer, and in them the drift is, if anything, slower while a carrier is in view (median −0.2 mm s^{-1} against 1.5 mm s^{-1} for ORB-SLAM3, one run of four faster in view). The filtered ORB-SLAM3 group is the only one large enough to test: there the matched-feature count is lower while traffic is in view in eleven of sixteen runs ($p = 0.05$) and the drift rate is higher in seven of ten ($p = 0.08$), with no change in the tracking-loss share. Taken together these suggest, and do not establish, that the class mask costs features when a carrier is in view and that the cost is paid in drift; the unfiltered estimators show no such cost.

The one large degradation of the realistic world is not a traffic encounter. The VINS-Fusion error on `realistic_002`, 1.8 m whole-run ATE, is a single 3 m step taken between 29 s and 33 s, at the end of

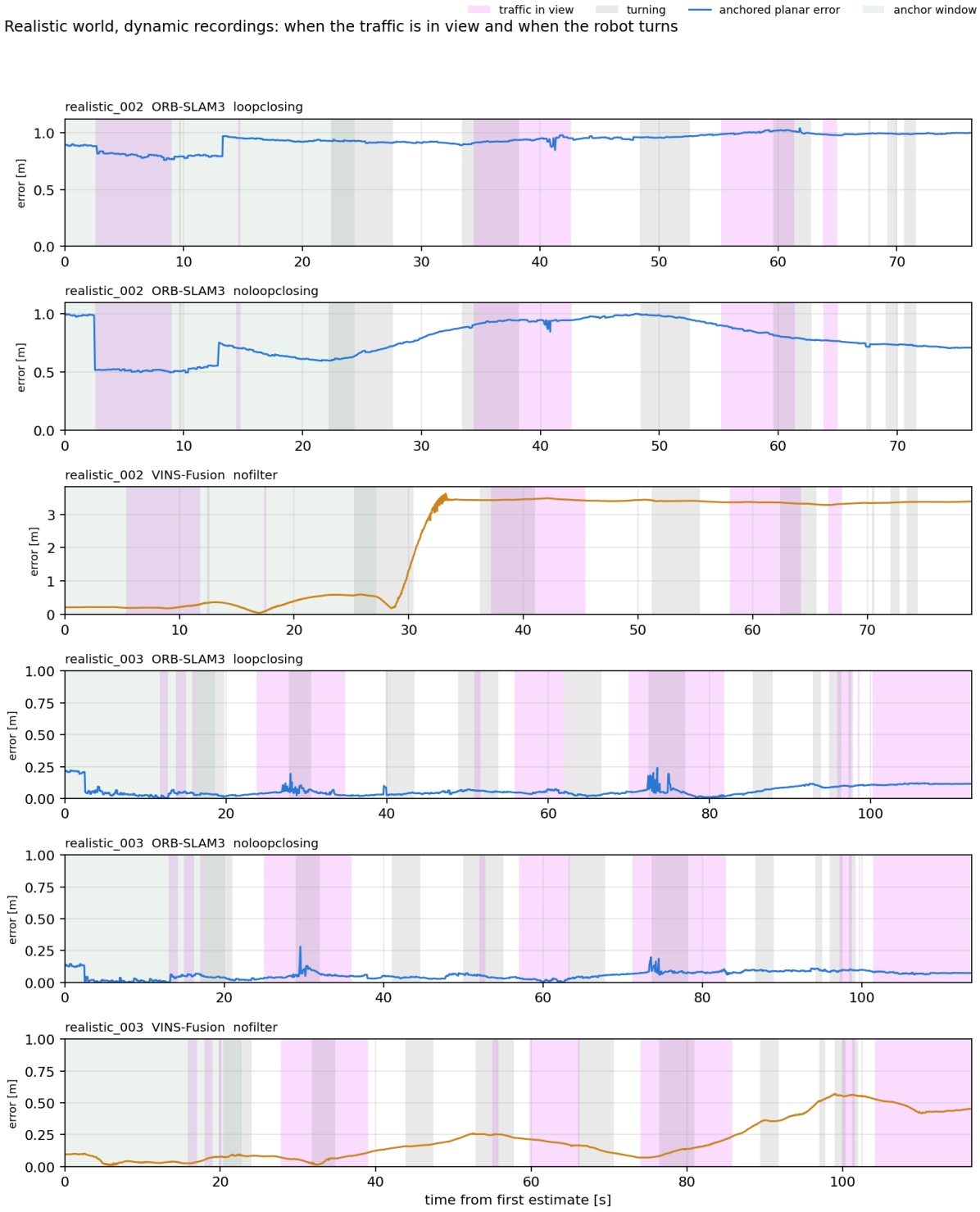


Figure I.4: Realistic dynamic recordings, unfiltered runs: anchored planar error (ground truth at the IMU point) with the moments a carrier or carried load is in the camera's view (violet) and the moments the robot turns (grey). The green band is the alignment window.

the first heading change and with no carrier in the camera's view (figure I.4, third panel); the carrier that passed at 5–12 s left no trace. The same bag gives both ORB-SLAM3 runs their 0.5–1.0 m step at 2.5 s and 13 s, the moments that same carrier enters and leaves the view. Whether the step is the estimator's or the ground truth's is not settled by these data.

[Coverage of the traffic measurement] Walking people are gz-sim actors and do not appear on the pose topic, so their visibility is unmeasured; occlusion is tested against shelf footprints only; and the carried loads move only while carried. The in-view state is therefore a lower bound on the moments something moving was in the image.

6 Insight 5: real robot (`dataset_realv6_000`): ORB-SLAM3 tracks the figure-of-eight to 0.4 m, VINS-Fusion diverges at both replay rates on a stale gravity norm, and one tracker sample decided the headline

The first real recording with a usable optical-tracker reference since the camera remount gives ORB-SLAM3 a provisional planar ATE of 0.41 m over 218 s at both replay rates, no tracking loss after its initialisation resets, and two VINS-Fusion runs that diverge to kilometres from the same moment at both replay rates, with an accelerometer bias frozen at the difference between the configured gravity norm and the gravity this IMU measures. Tables I.8 to I.11 and figures I.5 and I.7 to I.9 carry the evidence.

The recording. `dataset_realv6_000` was recorded on 31 October 2026 on the Gensurv floor: 220 s, 6 601 stereo pairs, the RealSense IMU at 200 Hz, wheel-speed feedback, and a figure-of-eight route of about 27 m at a median 0.15 m s^{-1}. Four estimator runs exist: ORB-SLAM3 and VINS-Fusion each at replay rates 1.0 and 0.5, ORB-SLAM3 on `config/wil/stereo_imu.yaml` and VINS-Fusion on the 31 October self-calibrated extrinsics (`config/wil_311026_selfcal`), all without a detector, so unfiltered. No filtered run exists, so the filter hypothesis remains untested on the real robot; what this slot can now say is how the unfiltered estimators behave on real imagery against an independent reference.

Reporting rules. Scoring is `script/eval_real_gt.py` unchanged: time offset from the cross-correlation of the tracker's speed with the bag's wheel speed, rigid SE(2) alignment with scale fixed, lever arm from the tracker marker to the IMU unmeasured and set to zero, so every number is *provisional*. The tool's validity rules were fixed before the data were read; the tracker export fails two of them (gaps above 1 s, a step above 0.5 m), so the tool labels every run *failed (ground truth invalid)* and the numbers below are reported as diagnostics, not as a result-slot headline. Outputs: `output/compare/real/realv6*/` and `output/compare/real/realv6-insights/`.

Table I.8: `dataset_realv6_000`: the same ORB-SLAM3 runs scored against the tracker export as delivered and with one sample removed (the 5.2 m excursion at 184.9 s). Time offset by wheel-speed cross-correlation with its peak correlation r; aligned planar ATE; path ratio = estimate / truth. The cleaned truth still fails the gap rule (66 gaps above 1 s, tracker at 1.7 Hz), so every value is provisional.

Run	truth	GT path [m]	offset [s]	r	ATE [m]	median [m]	path ratio
ORB-SLAM3, replay 1.0	raw	36.8	4.38	0.10	1.183	0.836	0.68
ORB-SLAM3, replay 1.0	cleaned	27.1	-4.14	0.61	0.409	0.343	0.93
ORB-SLAM3, replay 0.5	raw	36.8	4.38	0.10	1.232	0.889	0.70
ORB-SLAM3, replay 0.5	cleaned	27.1	-4.14	0.61	0.570	0.435	0.92

One tracker sample changed every headline. The export runs at 1.7 Hz, not the 7 Hz of the earlier recordings, with 66 gaps longer than 1 s, and it contains a single sample 5.2 m off the path. That sample added 10 m to the true path length (37.3 against 27.1 m), dominated the speed cross-correlation so that the

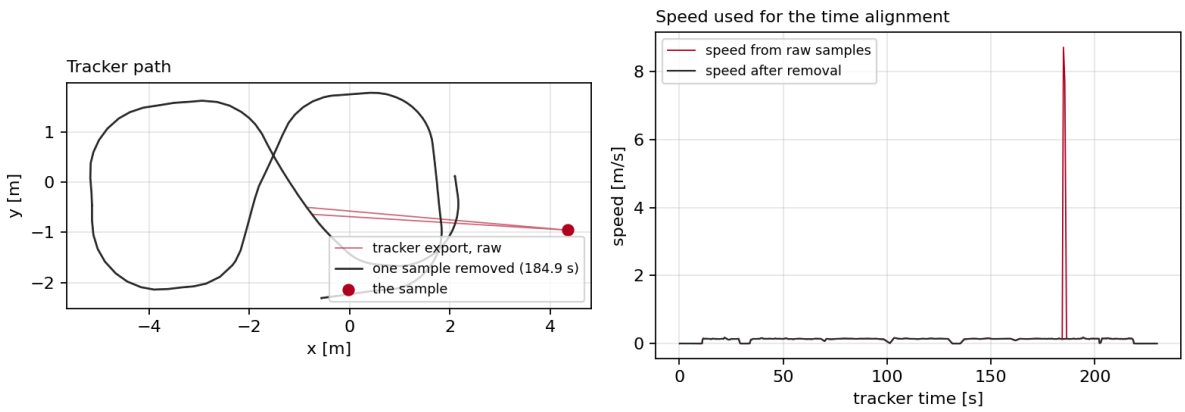


Figure I.5: The tracker export of `dataset_realv6_000`: one sample at 184.9 s lies 5.2 m off the path and back (left), which puts an 8.7 m s^{-1} spike into the speed profile the time alignment correlates against (right).

time offset came out as 4.4 s with a peak correlation of 0.10, and left ORB-SLAM3 with an ATE of 1.18 m and a path ratio of 0.68 (table I.8, rows "raw"). Removing that one sample, and nothing else, moves the offset to −4.1 s at a correlation of 0.61, the path ratio to 0.93 and the ATE to 0.41 m. The sparse rate and the gaps remain, so the cleaned truth still fails the gap rule; the cleaning is recorded here as a one-sample edit with its effect shown, not applied silently.

Table I.9: ORB-SLAM3 on `dataset_realv6_000` against the cleaned tracker truth, both replay rates. Aligned planar ATE after a rigid SE(2) fit with scale fixed; RPE at 1 s translational only; resets are ORB-SLAM3's `local_mapping_bad_imu` map resets, all inside the first 13 s; dropped frames are camera frames the node's queue discarded. Lever arm tracker-marker to IMU unmeasured (zero applied); the least-squares fit is a diagnostic only.

Run	matched	ATE [m]	median [m]	max [m]	$RPE_{1\,s}$ [m]	path ratio	resets	dropped frames
ORB-SLAM3, replay 1.0	247/247	0.409	0.343	0.837	0.073	0.93	2	3390 of 6476
ORB-SLAM3, replay 0.5	248/248	0.570	0.435	1.195	0.056	0.92	3	619 of 6560

Table I.10: ORB-SLAM3 on `dataset_realv6_000`, cleaned truth: growth rate of the planar error while the aligned estimate's heading changes faster than 10 ° s^{-1} and on straights, with the time spent in each state. Positive = error growing. One run per rate, so no test is possible.

Run	turning [mm/s]	straight [mm/s]	time turning [s]	time straight [s]
ORB-SLAM3, replay 1.0	-3.0	-0.4	11	193
ORB-SLAM3, replay 0.5	-44.1	-0.6	16	188

ORB-SLAM3 tracks the whole route at both replay rates. After two (rate 1.0) or three (rate 0.5) map resets during the first 13 s, while the inertial initialisation settles, tracking never leaves the OK state again; the error is a slow wave between 0.05 and 0.85 m with its crest around 120–140 s and its trough at 195 s, not a drift that accumulates (figure I.8). Turning does not raise it: the error shrinks slightly during the heading changes in both runs (table I.10), the same answer the simulation gave once the lever arm was removed. Real time costs frames but not accuracy here: at rate 1.0 the node dropped 3 390 of 6 476 frames and scored 0.41 m; at rate 0.5 it dropped 619 and scored 0.57 m, with the same shape of error curve.

What the 0.4 m is made of is not settled. The loops of the estimate are a uniform 7 % smaller than the tracker's (figure I.6; Sim(2) scale 1.09; path ratio 0.93), which a stereo-baseline error, a tracker scale error or the unmeasured lever arm could each produce; the least-squares lever-arm fit (0.31 m behind the IMU,

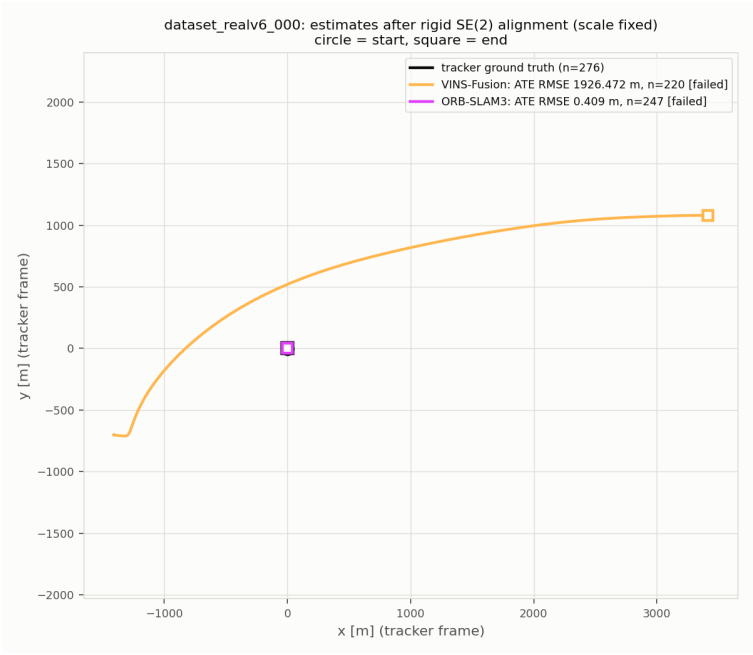


Figure I.6: ORB-SLAM3 at replay rate 1.0 against the cleaned tracker truth after rigid SE(2) alignment. The estimate traces the figure-of-eight with its loops about 7 % smaller than the tracker's (Sim(2) scale diagnostic 1.09, not applied).

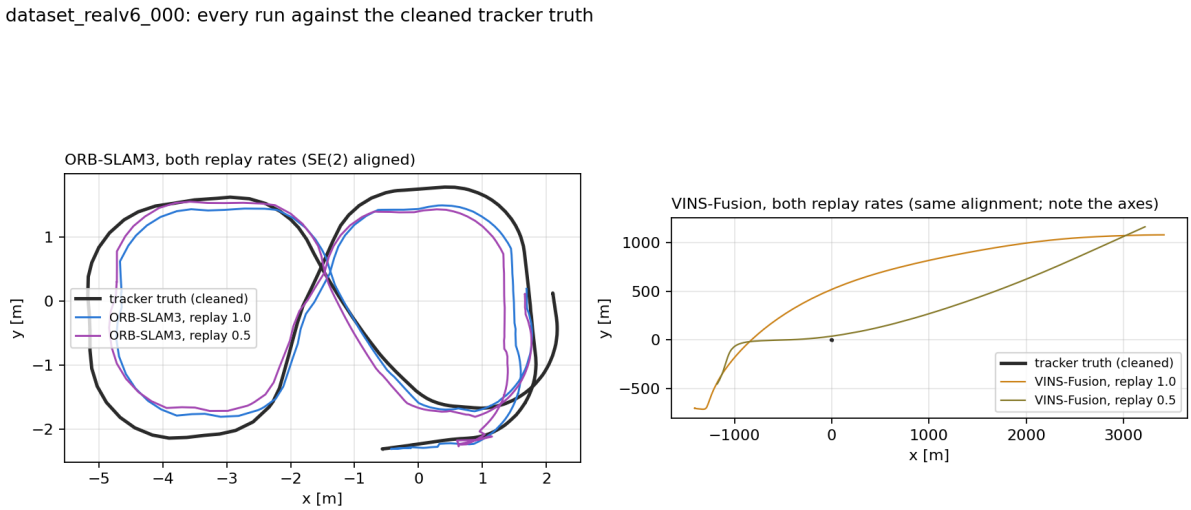


Figure I.7: All four runs against the cleaned tracker truth after the same rigid SE(2) alignment. Left: ORB-SLAM3 at replay 1.0 and 0.5, both on the route. Right: VINS-Fusion at both rates, which leave the 10 m floor within a minute; note the axis scale.

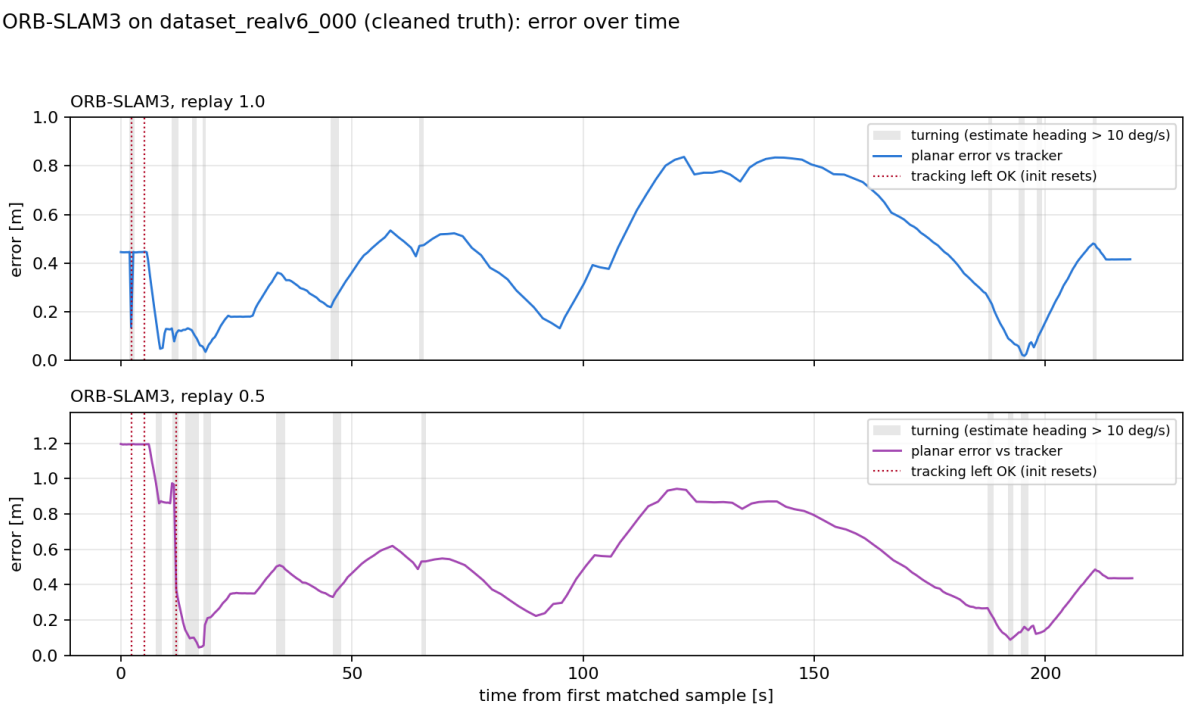


Figure I.8: ORB-SLAM3 planar error against the cleaned tracker truth over the run, both replay rates. Grey bands: the aligned estimate's heading changes faster than 10 ° s^{-1}. Dotted red: the frames at which tracking left the OK state, all inside the first 13 s (the `local_mapping_bad_imu` initialisation resets).

not applied) would bring the ATE to 0.31 m, which is the size of the effect the unmeasured arm can hide.

Table I.11: VINS-Fusion on `dataset_realv6_000` at both replay rates: camera frames the node received (bag: 6 601), median interval between processed frames, the time at which the estimated speed first exceeds 1 m s^{-1} (the robot never exceeds 0.9 m s^{-1}), the distance of the estimate from its origin at three times, and the accelerometer-bias magnitude the estimator settled on between 30 and 50 s. The configured gravity norm is 9.2642; the IMU in this bag reads 9.886 at rest.

Run	frames	interval [s]	speed >1 at [s]	$\|p\|$ 60 s [m]	$\|p\|$ 120 s [m]	$\|p\|$ end [m]	accel. bias [m/s^2]
VINS-Fusion, replay 1.0	1047	0.20	38	10	417	5164	0.68
VINS-Fusion, replay 0.5	2264	0.07	37	57	523	4714	0.72

VINS-Fusion diverges at both replay rates from the same moment, so the frame rate is not what breaks it. The two nodes received 1 047 and 2 264 of the bag's 6 601 frames (one in six and one in three), initialised at 0.33 s and never raised a reset; in both the estimated speed passes 1 m s^{-1} at 37–38 s, while the robot never exceeds 0.9 m s^{-1}, and from there the position runs away to about 5 km (table I.11, figure I.9). The quantity that identifies the cause is the accelerometer bias: both runs settle it at 0.68–0.72 m s^{-2} and then hold it, which is within 0.1 m s^{-2} of the gap between the gravity norm the configuration fixes, 9.2642, measured on an August recording of an earlier IMU, and the 9.878 m s^{-2} this bag's IMU reads at rest. The estimator absorbs the mismatch into the bias while the visual constraints are strong, and once they thin out the mis-scaled gravity integrates into velocity. ORB-SLAM3 on the same bag is unaffected because its launch rescales the accelerometer (`imu_accel_scale`) instead of fixing a gravity norm. The frame starvation is real and secondary: it weakens the visual constraint that was holding the bias in place, and it sits on VINS-Fusion's input path, since the ORB-SLAM3 node received 6 476 frames from the same bag.

[What this slot still needs before it is a result] A tracker export at its nominal 7 Hz without gaps, or an explicit rule for scoring on a sparse reference; the marker-to-IMU lever arm measured, which bounds 0.1 m of the ORB-SLAM3 figure; a VINS-Fusion run with `g_norm` set to what this IMU measures (or

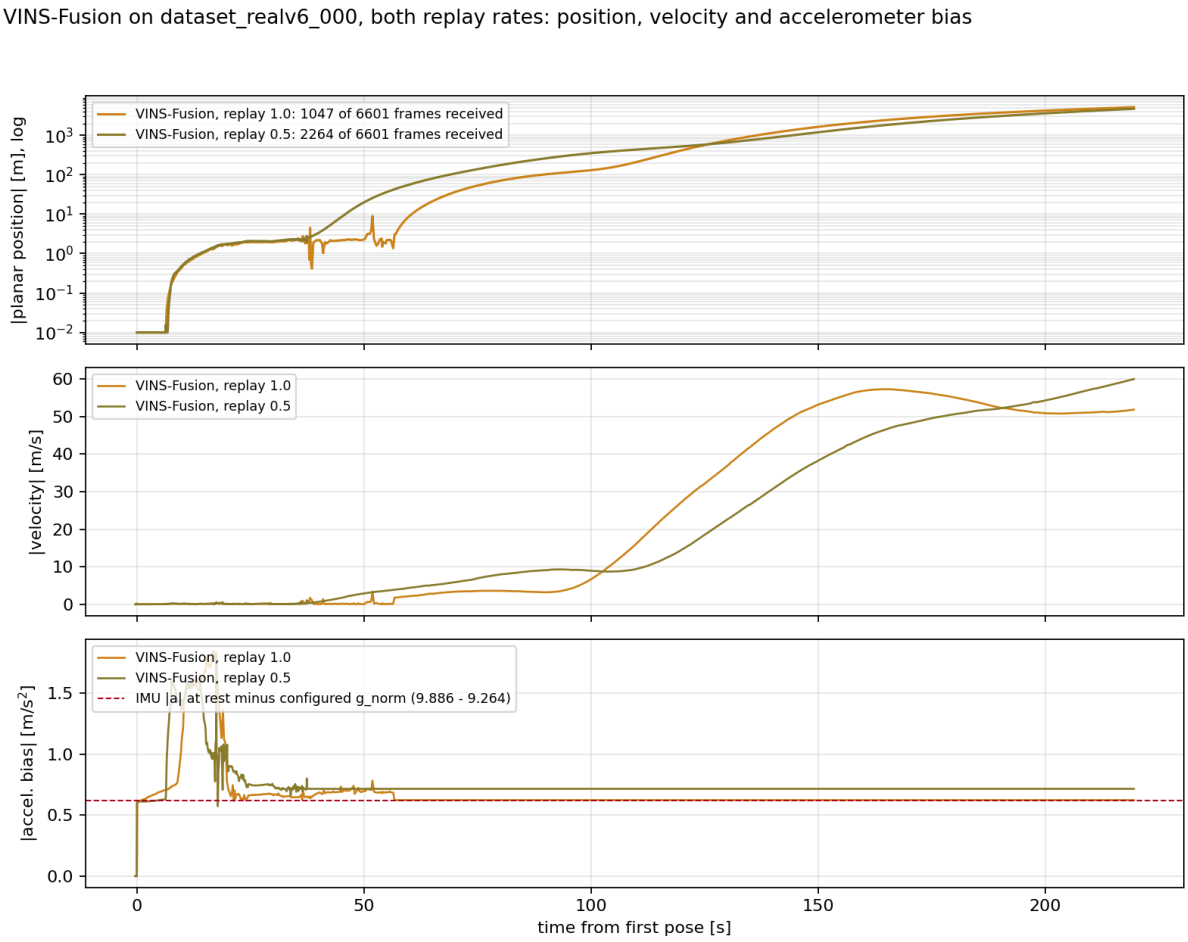


Figure I.9: VINS-Fusion on `dataset_realv6_000` at both replay rates: distance of the estimate from its origin (log scale), estimated speed, and the accelerometer-bias magnitude the estimator holds. The dashed line is the difference between what the IMU measures at rest in this bag (9.878 m s^{-2}) and the gravity norm the configuration fixes (9.2642).

the accelerometer rescaled as ORB-SLAM3 does) and with the full frame stream delivered; and at least one filtered run per estimator, without which Experiment 3's hypothesis has no real-robot test.

7 What these data cannot answer

- **Whether the filter helps.** No admissible filtered-versus-unfiltered pair exists in either simulated testbed, and no filtered run exists on the real robot; nothing here changes that.
- **Why VINS-Fusion fails in the extreme world.** The failure is immediate and not preceded by feature loss or by a rise in moving-object area; the mechanism is not identified.
- **The traffic effect with confidence.** Two to four unfiltered runs per estimator in the realistic world, with the people unmeasured, bound the effect rather than estimate it.
- **The feature-collapse signal, robustly.** The onset threshold is three times the static median error, and that median carries the lever arm. Recomputing the thresholds at the IMU point lowers them, moves the onsets earlier, and weakens the ORB-SLAM3+YOLO feature collapse of figure I.3 to a dip to about half in four runs (`fig_onset_epoch_lever0p338_light.png`). The collapse is real at the chapter's definition and threshold-dependent in size.

8 Summary

Consequences for Chapter 4. Re-score every simulation ATE with the ground truth moved to the IMU point before it is quoted, in Experiments 1 and 3. Report the extreme world as an immediate-failure testbed in which the filter cannot be evaluated, not as a drift testbed. Report the realistic world's traffic effect as bounded, not measured, and the class mask's feature cost as the one mechanism these data show. For the real-robot slot, quote ORB-SLAM3's 0.41 m as provisional with the sparse-truth and lever-arm caveats attached, report the VINS-Fusion failure as a configuration defect (stale gravity norm) compounded by frame delivery rather than an estimator result, and keep the slot *partial* until a filtered run and a full-rate truth exist.

Table I.12: Questions, answers and the strength of the evidence.

	Answer	Evidence	Confidence
Q5 lever arm	Yes: 0.34 m of forward offset is in every simulation ATE	minimum of the sweep at the SDF value in 30 runs, two estimators, two testbeds	high
Q1 turning	No, once the lever arm is removed	contrasts of tables I.3 and I.4; heading error flat	medium (small groups)
Q2 object in view, extreme	Not measurable within a bag; moving area does not modulate tracking loss or jumps	table I.6	medium
Q2 traffic in view, realistic	No effect on the unfiltered drift; the class mask loses features and gains drift while traffic is in view (suggestive)	table I.7, figure I.4	low (n)
Q3 features before onset	ORB-SLAM3+YOLO yes (collapse to one fifth); VINS-Fusion no	figure I.3	medium
Q4 state at onset	On the first straight, not turning; objects in view	table I.5	high
Real robot, ORB	Tracks the whole 218 s route; provisional ATE 0.41 m; no turning effect; 7 % scale shortfall unexplained	table I.9 and figure I.8	medium (provisional, sparse truth)
Real robot, VINS	Diverges at both rates from 37 s; accelerometer bias frozen at the 0.6 gap between configured gravity norm and the IMU's; frame starvation secondary	table I.11, figure I.9	high (mechanism), medium (attribution)
Real robot, truth	One tracker sample set the raw ATE to 1.18 m; truth at 1.7 Hz fails the gap rule	table I.8	high